

Smart Internship Management System

Group Members:

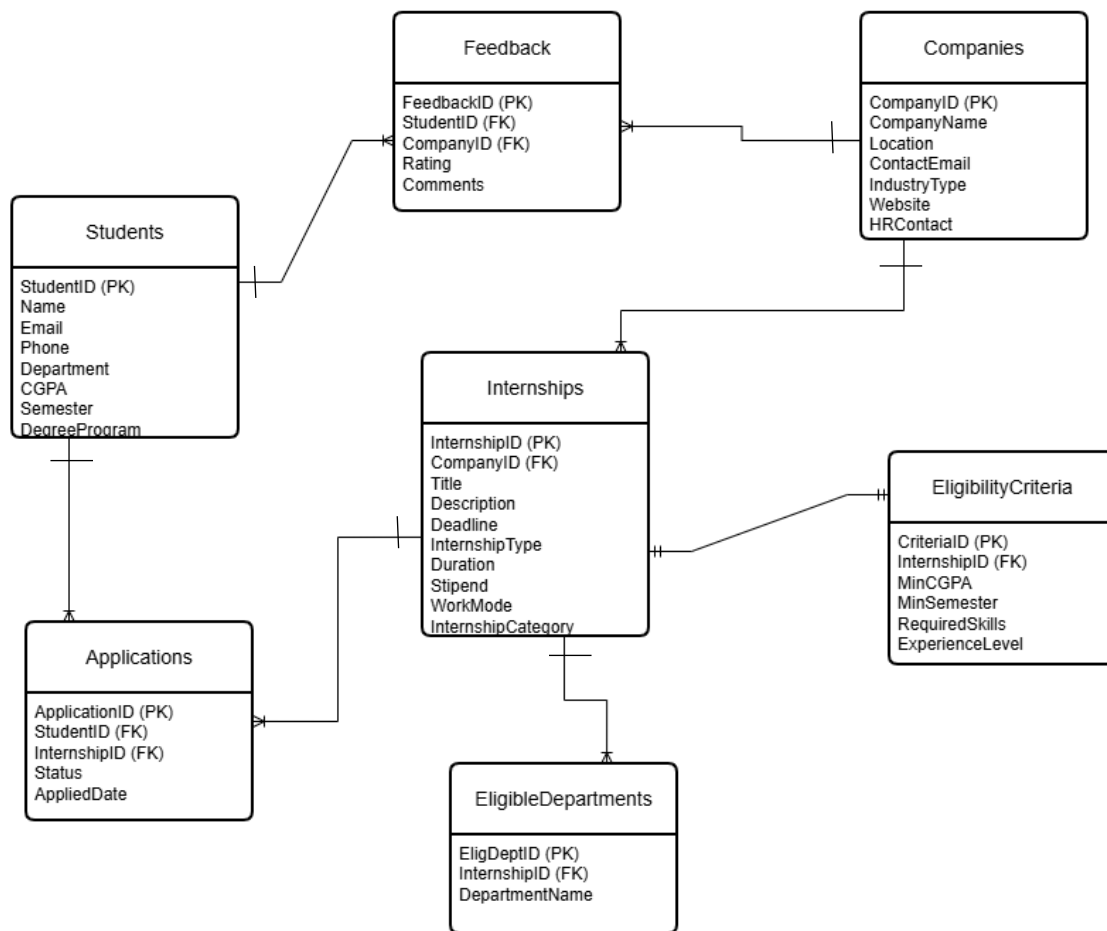
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Milestone Tracking:

Milestones	Date	Version	Remarks
ERD Diagram and Schema	24/4/2026	1.0	
ERD Diagram and Schema (Updated)	10/5/2026	1.1	

Entity Relationship Diagram (ERD):



Relational Database Schema Description:

1. Students Table

Attributes:

- StudentID (Primary Key)
- Name
- Email
- Phone

- Department
- CGPA
- Semester
- DegreeProgram

2. Companies Table

Attributes:

- CompanyID (Primary Key)
- CompanyName
- Location
- ContactEmail
- IndustryType
- Website
- HRContact

3. Internships Table

Attributes:

- InternshipID (Primary Key)
- CompanyID (Foreign Key → Companies)
- Title
- Description
- Deadline
- InternshipType (e.g., Paid / Unpaid / Academic Credit)
- Duration (in weeks)
- Stipend
- WorkMode (Remote / Onsite / Hybrid)
- InternshipCategory (e.g., Engineering, Design, Marketing)

4. Applications Table

Attributes:

- ApplicationID (Primary Key)
- StudentID (Foreign Key → Students)
- InternshipID (Foreign Key → Internships)
- Status (Pending / Accepted / Rejected)
- AppliedDate

5. Feedback Table

Attributes:

- FeedbackID (Primary Key)
- StudentID (Foreign Key → Students)
- CompanyID (Foreign Key → Companies)
- Rating
- Comments

6. EligibilityCriteria Table

Attributes:

- CriteriaID (Primary Key)
- InternshipID (Foreign Key → Internships)
- MinCGPA (e.g., 2.5)
- MinSemester (e.g., 5th semester or above)
- RequiredSkills (e.g., Python, SQL, Communication)
- ExperienceLevel (None / Beginner / Intermediate)

7. EligibleDepartments Table

Purpose: Junction table linking internships to their eligible student departments. An internship may be open to multiple departments, and each department may be eligible for multiple internships (M:N relationship).

Attributes:

- EligDeptID (Primary Key)
- InternshipID (Foreign Key → Internships)
- DepartmentName (e.g., BSCS, BSSE, BSIT)

Relationships:

- One **Student** can submit many **Applications** (1:M)
- One **Internship** can receive many **Applications** (1:M)
- One **Company** can post many **Internships** (1:M)
- One **Student** can give many **Feedback entries** (1:M)
- One **Company** can receive many **Feedback entries** (1:M)

- One **Internship** has exactly one **EligibilityCriteria** record (1:1)
- One **Internship** can be open to many **Departments**; one Department can be eligible for many Internships — resolved via **EligibleDepartments** junction table (M:N)

Updated ERD Structure Summary:

The following entities and relationships are reflected in the ERD diagram above (v2.0). New additions are marked with [NEW] and modified entities with [UPDATED].

- **[UPDATED] Students** — Added: CGPA, Semester, DegreeProgram
- **[UPDATED] Companies** — Added: IndustryType, Website, HRContact
- **[UPDATED] Internships** — Added: InternshipType, Duration, Stipend, WorkMode, InternshipCategory
- **[NEW] EligibilityCriteria** — Stores per-internship eligibility rules: MinCGPA, MinSemester, RequiredSkills, ExperienceLevel. Linked 1:1 to Internships.
- **[NEW] EligibleDepartments** — Junction table resolving the M:N relationship between Internships and eligible student departments.

Eligibility Checking Logic:

When a student submits an application, the system verifies eligibility by comparing the student's profile against the internship's EligibilityCriteria record and EligibleDepartments entries. An application is considered eligible only if all of the following conditions are satisfied:

- Student.CGPA $\geq$ EligibilityCriteria.MinCGPA
- Student.Semester $\geq$ EligibilityCriteria.MinSemester
- Student.Department EXISTS IN EligibleDepartments for that InternshipID

If any condition fails, the Application.Status is automatically set to “Ineligible” before submission, preventing invalid applications from reaching the company.